

Order-Separating Exponents for Classical Integer Factorization: Two Restricted Theorems and Six Structural Barriers

MOSEF Research Program

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Abstract

We study order-separating exponent families for classical integer factorization, measuring complexity in the binary input length. We first give exact support and prime-power valuation semantics for candidates of the form $\gcd(g^d - 1, N)$, including every failure branch. On a factor-dependent but explicit hereditary semismooth-asymmetry promise class, we prove that fresh uniform bases yield a Las Vegas complete-factorization algorithm with polynomial expected bit complexity and at least $5/12$ split probability at each complete schedule cycle at a promised residual node. We also prove an exact Lucas root-count formula and use fresh uniform, independently parameterized Lucas values to obtain a second Las Vegas theorem on a hereditary $p + 1$ -asymmetry promise, with at least $1/12$ split probability at a witness trial. We prove six structural barriers. Divisor coverage does not imply multiplicative-order separation; the exact missing condition is injectivity of divisibility signatures. A natural Lucas parameter map $P = a + a^{-1}$ is also not an independent group channel: $V_d(P, 1) - 2 = a^{-d}(a^d - 1)^2$, so it cannot enlarge the multiplicative family's success domain when exponent 2 is retained and can worsen prime-power collisions. For one common finite exponent schedule, we characterize the union of the local $p - 1$ and $p + 1$ promises by a combined prime signature and prove a finite-distribution upper bound plus a magnitude-conditioned zero-density corollary for balanced semiprimes. These signatures are factor-dependent analytical objects, not a recognizer. Finally, an exact divisor budget shows that a polynomial-size explicit schedule whose maximum exponent length is $o(k \log k)$ hits at most $2^{o(k)}$ odd primes; on any stipulated exponentially large common-input-length prime population, its promised-pair fraction vanishes. This last result makes no prime-distribution claim. We then formalize one compressed-representation case: a factor-oblivious multiplication-only straight-line program. A t -node program realizes formal exponent at most 2^t , so $t = o(k \log k)$ still falls under the explicit-list barrier, and the short descriptor 2^{2^s} needs exactly 2^s multiplication nodes in this model. This is not a lower bound for equal-residue computation at a fixed modulus or for broader algebraic representations. At the exact boundary, an optimized divisor budget gives $\log_2 \tau(d) \leq (1 + o(1))\ell(d)/\log_2 \ell(d)$. Hence an exponent-length coefficient c still yields at most $2^{ck+o(k)}$ hits and a vanishing fraction on any stipulated $2^{\alpha k}$ -size population when $c < \alpha$. First-primes primorials have exponential divisor capacity at this boundary, but no proved asymptotic $d \pm 1$ prime yield; repeated squaring shows that boundary node count alone is insufficient. Deterministic bounded searches and independent Python, Rust, and C# implementations audit these statements but are not used as proofs. No general polynomial-time classical factoring algorithm or lower bound is claimed.

1 Introduction and contributions

Let N be a positive integer represented in binary. A recurring strategy for factoring N is to choose an algebraic element, raise it to a structured exponent, and take a greatest common divisor with N . The difficulty is not merely to hit an unknown prime-factor order. A useful exponent must hit

some prime-power components without hitting all of them, and its bases, exponents, representation, construction, evaluation, and recursion must all cost polynomially many bit operations in $\log N$.

This paper treats the proposed Multi-Group Order-Separating Exponent Factoring (MOSEF) direction as a falsifiable research program rather than a presumed algorithm. Its contributions are:

- exact support and capped-valuation criteria for multiplicative candidates, including the prime-power obstruction to an all-composite support-only definition;
- an unconditional Las Vegas complete-factorization theorem on a precisely stated hereditary semismooth-asymmetry promise class, with correctness, termination, sampling, recursion, and bit-complexity proofs;
- an exact Lucas root-count formula and a second unconditional Las Vegas theorem on a hereditary nonsplit $p + 1$ -asymmetry promise, using fresh factorization-independent parameters;
- a coverage-separation barrier showing that nonempty divisibility signatures are insufficient and that universal separation of distinct order profiles is exactly signature injectivity;
- a channel-correlation barrier showing that the natural conjugate Lucas map only squares multiplicative valuations and therefore does not create an independent $p + 1$ channel;
- a self-contained common-schedule finite-distribution upper bound for the union of the local $p - 1$ and $p + 1$ promises, including a magnitude-conditioned zero-density corollary for balanced semiprimes;
- an explicit-list exponent-encoding barrier: a polynomial-size schedule of $o(k \log k)$ -bit exponents has a subexponential global divisor-hit set and vanishing promised-pair fraction on every stipulated exponentially large common-input-length prime population;
- a multiplication straight-line compression barrier: exact formal exponent realization grows by at most one bit per charged multiplication, so short symbolic syntax alone does not evade the explicit-list result; and
- a constant-sensitive exact-boundary barrier, together with an explicit $\Theta(k \log k)$ -cost primordial capacity family and a repeated-squaring counterexample to sufficiency of boundary node count; and
- deterministic falsification searches, canonical vectors, and cross-language checks whose finite scope and hashes are recorded explicitly.

The positive theorems are restricted: membership in their promise classes can depend on the unknown factorization and is not recognized by the algorithms. The six barriers are also scoped: one concerns explicit order signatures, one concerns only $P = a + a^{-1}$, and two concern a common explicit finite schedule under separate magnitude or encoding hypotheses. The fifth concerns only formal exponents in a factor-oblivious multiplication-only DAG. The sixth exposes a coefficient threshold for stipulated populations but is silent at and above the critical coefficient. These boundaries are part of the results.

Section 2 records the imported complexity landscape. Sections 3 and 4 define candidate semantics and the recursive framework. Section 5 gives the first restricted theorem. Sections 6 and 7 prove two barriers, Section 8 gives the nonsplit Lucas theorem, Section 9 gives the combined-promise density barrier, Section 10 gives the exponent-encoding barrier, Section 11 gives the multiplication

straight-line compression barrier, and Section 12 treats the exact boundary coefficient. Later sections separate algorithms, experiments, and observations. Complete proofs and reproduction details appear in the appendices.

2 Related work and complexity landscape

EXT-001 [PROVED, IMPORTED]. Agrawal, Kayal, and Saxena give an unconditional deterministic primality test running in time polynomial in the binary input length [1]. This is a decision theorem and does not itself find a nontrivial factor.

EXT-002 [PROVED, IMPORTED]. Lenstra and Pomerance prove a probabilistic complete-factorization algorithm with expected running time $L_N[1/2, 1 + o(1)]$ [2]. This is rigorous and subexponential, but it is not polynomial in $\log N$. Lenstra’s elliptic-curve method motivates varying group orders; the specific expected time expression highlighted in its abstract is explicitly conjectural [3], so we use it only as motivation.

EXT-003 [CONDITIONAL, IMPORTED]. Umans and Wang’s Theorem 5.5, assuming their Strong Prefactored Divisor Conjecture, gives deterministic integer-factorization time

$$\tilde{O}\left(N^{\max(\alpha, \beta)/2 + o(1)}\right)$$

[5]. This remains exponential in the binary input length. Their integer mechanism covers actual prefactored divisors and has a separate collision-resolution process; it does not claim that divisor coverage implies multiplicative-order separation.

EXT-004 [PROVED, IMPORTED]. Williams states and uses, with attribution to Lehmer, Lucas congruences for sequences with $\Delta = P^2 - 4Q$. For an odd prime $p \nmid Q$, $(\Delta/p) = \varepsilon$, and $s > 0$, the imported Lucas congruences give

$$U_{(p-\varepsilon)s}(P, Q) \equiv 0 \pmod{p}, \quad V_{(p-\varepsilon)s}(P, Q) \equiv 2Q^{s(1-\varepsilon)/2} \pmod{p}.$$

For $Q = 1$, $\varepsilon = -1$ is the nonsplit $p + 1$ branch and $\varepsilon = +1$ the split $p - 1$ branch [6]. The zero-discriminant prime component is outside this statement.

EXT-005 [PROVED, IMPORTED]. Rosser and Schoenfeld’s Theorem 3 corollary gives, for the n -th prime,

$$p_n < n(\log n + \log \log n) \quad (n \geq 6),$$

with natural logarithms [4]. We use only the consequence $p_n = O(n \log n)$ to cost the first-primes primorial constructor; the source supplies no $d \pm 1$ prime-yield statement.

3 Model and definitions

DEF-001 [DEFINITION]. The input length is

$$m = \lceil \log_2 N \rceil.$$

A bound polynomial in N , $\sqrt{N}$, or an enumerated group size is not therefore polynomial in the input length.

DEF-002 [DEFINITION]. Let $P(N)$ be the set of distinct prime divisors of N . For $d > 0$ and $\gcd(g, N) = 1$, define

$$D_N(g, d) = \{p \in P(N) : \text{ord}_p(g) \mid d\}.$$

The pair (g, d) is a support order separator when this set is nonempty and proper.

DEF-003 [DEFINITION]. If

$$N = \prod_{p \in P(N)} p^{e_p},$$

define the capped valuation profile

$$a_p(N, g, d) = \min\{e_p, \nu_p(g^d - 1)\},$$

with $\nu_p(0) = +\infty$. A valuation separator has some $a_p > 0$ and some $a_q < e_q$, where $p = q$ is allowed.

A candidate polynomial order-separating family must construct explicit base and exponent sets without access to the factorization, use polynomially many members, bound every exponent's bit length, evaluate every required group operation in polynomial bit complexity, and guarantee a valid split at every recursive composite in its claimed domain.

3.1 Open and refuted universal targets

OPEN-001 [REFUTED]. The original support-only requirement cannot cover every composite input. For $N = p^e$, $e \geq 2$, the set $P(N)$ is a singleton, so a support is either empty or all of $P(N)$, never nonempty and proper. This does not prevent a partial prime-power valuation from producing a factor.

OPEN-002 [OPEN]. It remains unknown here whether deterministic polynomial-time primality and exact perfect-power preprocessing can be followed by a factorization-independent support-POSF constructor that succeeds on every remaining composite recursive cofactor with polynomial representation, construction, exponent, evaluation, and recursion bounds.

OPEN-003 [OPEN]. It also remains unknown here whether a factorization-independent valuation-separating family satisfying the same bounds exists for every composite input, including prime powers.

4 Order separators and algorithmic framework

LEM-001 [PROVED]. If $D_N(g, d)$ is nonempty and proper, then

$$1 < \gcd(g^d - 1, N) < N.$$

The complete proof is in Appendix A.1.

LEM-002 [PROVED]. The exact GCD is

$$\gcd(g^d - 1, N) = \prod_{p \in P(N)} p^{\min\{e_p, \nu_p(g^d - 1)\}}.$$

It is a proper factor exactly when the capped valuation profile is neither all zero nor full. For square-free N , this is equivalent to the support condition; for $N = p^e$, it is equivalent to $0 < \nu_p(g^d - 1) < e$. Appendix A.2 gives the proof.

One candidate is evaluated as follows.

```
EVALUATE-CANDIDATE(N, g, d):
  require N >= 2 and d > 0
  g <- g mod N
```

```

u <- gcd(g, N)
if 1 < u < N: return DIRECT-FACTOR(u)
if u = N: return INVALID-BASE
r <- modular_power(g, d, N)
h <- gcd((r - 1) mod N, N)
if h = 1: return MISS
if h = N: return SIMULTANEOUS-COLLISION
return FACTOR(h)

```

A complete wrapper first recognizes prime inputs, expands exact maximal perfect powers, and invokes a family only on residual composites. It validates every reported divisor before recursively factoring both children. `UNRESOLVED` is never interpreted as prime or as proof of nonmembership.

Let $C(m)$ bound construction, $B(m)$ the number of canonical residue bases, $E(m)$ the number of exponents, and $L(m)$ their maximum bit length. With multiplication cost $M(m)$ and GCD cost $G(m)$, one explicit Cartesian-product pass costs

$$O(C(m) + B(m)E(m)(M(m)L(m) + G(m))).$$

This is a conditional framework, not an available universal algorithm: all of C, B, E, L , representation conversion, preprocessing, and the success guarantee on every recursive cofactor must be polynomial.

5 A restricted semismooth-order theorem

Let $B(k)$ and $R(k)$ be fixed nondecreasing, positive, polynomially bounded integer functions computable in time polynomial in k , and put

$$M_B(k) = \text{lcm}(1, 2, \dots, B(k)).$$

DEF-004 [DEFINITION]. A composite non-perfect-power K , with $k = \lceil \log_2 K \rceil$, has a semismooth-asymmetry witness when distinct primes $p, q \mid K$ and $1 \leq t \leq R(k)$ satisfy

$$p - 1 \mid tM_B(k), \quad q - 1 \nmid tM_B(k).$$

An input belongs to the hereditary promise class when every divisor $K \mid N$ that is composite and not a perfect power has such a witness at its local bit length. Membership is promised, not recognized.

THM-001 [PROVED]. For every fixed schedule pair B, R above, there is a classical Las Vegas algorithm that completely factors every input in the hereditary semismooth-asymmetric promise class, is always correct, terminates with probability one, and has expected running time polynomial in the binary input length.

At a residual composite K , the algorithm repeatedly enumerates $1 \leq t \leq R(k)$. For every t , it draws a fresh exact uniform residue $a \bmod K$, first tries $\text{gcd}(a, K)$, and then tries

$$\text{gcd}(a^{tM_B(k)} - 1, K).$$

At a witness value, every unit is 1 after exponentiation modulo p , while at most half of the units are 1 modulo q . Proper nonunits already factor K . The one-trial success probability is therefore at least

$$\frac{K - 1 - \varphi(K)/2}{K} \geq \frac{K - 1}{2K} \geq \frac{5}{12}.$$

Fresh samples give a geometric tail. The exponent bit length is

$$O(B(k) \log B(k) + \log R(k)),$$

and the complete recursion has $O(m)$ nodes after exact maximal-power preprocessing. Appendix B.1 proves correctness, almost-sure termination, and expected polynomial bit complexity.

REF-001 [REFUTED]. Replacing fresh random bases by a fixed base is invalid. For

$$N = 51, \quad p = 3, \quad q = 17, \quad a = 2, \quad d = \text{lcm}(1, \dots, 8) = 840,$$

we have $p - 1 \mid d$ and $q - 1 \nmid d$, but $\text{ord}_{17}(2) = 8 \mid d$. Hence $\text{gcd}(2^{840} - 1, 51) = 51$, a simultaneous collision.

6 Difference coverage versus order separation

DEF-005 [DEFINITION]. For a finite positive exponent family Δ , define

$$\Sigma_{\Delta}(r) = \{d \in \Delta : r \mid d\}.$$

The family has the n -divisor property when every signature on $\{1, \dots, n\}$ is nonempty. It separates a profile $(r_1, \dots, r_s)$ when some $d \in \Delta$ is divisible by at least one but not all profile entries.

BAR-001 [PROVED]. Divisor coverage does not imply distinct-order separation, even for positive difference families. A family separates a profile exactly when the profile signatures are not all equal, and it separates every profile containing distinct orders in $\{1, \dots, n\}$ exactly when

$$r \mapsto \Sigma_{\Delta}(r)$$

is injective on that interval. Any explicit family with both nonempty and injective signatures satisfies

$$|\Delta| \geq \lceil \log_2(n + 1) \rceil.$$

If

$$\Delta = \Delta^+(S, T) = \{s - t : s \in S, t \in T, s > t\},$$

then also

$$|S||T| \geq |\Delta| \geq \lceil \log_2(n + 1) \rceil.$$

The smallest difference-family counterexample is

$$S = \{3\}, \quad T = \{1\}, \quad \Delta^+(S, T) = \{2\}.$$

Both orders 1 and 2 divide the sole difference. The collision occurs in the multiplicative mechanism itself:

$$\text{ord}_2(5) = 1, \quad \text{ord}_3(5) = 2, \quad \text{gcd}(5^2 - 1, 6) = 6.$$

A control construction realizes every difference from 1 through $\lceil \sqrt{n} \rceil^2$ but uses $\Theta(n)$ pairs, which is exponential when n has magnitude exponential in the input length. Appendix C.1 gives the complete characterization, lower bound, and construction proof.

REF-002 [REFUTED]. The implication “ n -divisor difference cover $\Rightarrow$ distinct-order separator” is false. This refutes only the attempted transfer into support-POSF semantics; it does not refute the conditional Umans–Wang integer algorithms.

7 A two-channel correlation barrier

DEF-006 [DEFINITION]. For $N \geq 2$, a unit $a \bmod N$, an arbitrary parameter P , and $d > 0$, define

$$G_M(N, a, d) = \gcd(a^d - 1, N),$$

$$G_\Delta(N, P) = \gcd(P^2 - 4, N), \quad G_L(N, P, d) = \gcd(V_d(P, 1) - 2, N),$$

where

$$V_0(P, 1) = 2, \quad V_1(P, 1) = P, \quad V_{j+1}(P, 1) = PV_j(P, 1) - V_{j-1}(P, 1).$$

Every GCD is classified as a miss, proper factor, or full collision. If $G_\Delta = N$, the sequence GCD is still separately classified; for example, $N = 15, P = 8, d = 1$ has a full discriminant GCD and a sequence factor 3.

BAR-002 [PROVED]. Under the factorization-independent conjugate map

$$P(a) = a + a^{-1} \pmod{N},$$

we have, for every $d \geq 0$,

$$V_d(P(a), 1) - 2 = a^{-d}(a^d - 1)^2 \pmod{N}$$

and

$$P(a)^2 - 4 = a^{-2}(a^2 - 1)^2 \pmod{N}.$$

The two sequence residues have identical prime support, and their raw GCDs agree for square-free N . If a finite exponent family contains 2, any proper conjugately derived sequence GCD implies a proper multiplicative GCD at the same exponent, while a proper discriminant GCD implies a proper multiplicative GCD at exponent 2. Adding the derived Lucas family therefore does not enlarge the multiplicative family's success domain.

The map may worsen a prime-power split:

$$(N, a, P, d) = (25, 2, 15, 4), \quad G_\Delta = 1, \quad G_M = 5, \quad G_L = 25.$$

Its discriminant is a square modulo every nondegenerate odd prime, so the map forces Williams's split branch instead of constructing a nonsplit $p + 1$ channel. Appendix D.1 supplies the complete recurrence, valuation, and family-success proof.

REF-003 [REFUTED]. The conjugate map does not produce an independent Lucas separator. This is map-specific. With independently selected P ,

$$(N, a, P, d) = (15, 2, 9, 3)$$

has $G_M = 1$, $G_\Delta = 1$, and $G_L = 5$. This proves that an arbitrary parameter can complement a miss; it does not prove an independent failure distribution or a universal construction.

8 A restricted nonsplit Lucas theorem

Let $B(k)$, $R(k)$, and $M_B(k)$ have the same fixed polynomial schedule conditions as in Section 5.

DEF-007 [DEFINITION]. An odd composite non-perfect-power K has a nonsplit Lucas asymmetry witness when distinct odd primes $p, q \mid K$ and some $1 \leq t \leq R(k)$, for $d = tM_B(k)$, satisfy

$$p + 1 \mid d, \quad q + 1 \nmid d.$$

The hereditary promise requires such a witness at every reached odd composite non-perfect-power divisor. It is nonempty: with $B(k) = R(k) = 2$, the input 15 has $p = 3, q = 5, t = 2, d = 4$. Membership may depend on the unknown factorization and is not recognized by the algorithm.

LEM-003 [PROVED]. For every odd prime q and $d > 0$,

$$\#\{P \in \mathbb{F}_q : V_d(P, 1) = 2\} = \frac{\gcd(d, q-1) + \gcd(d, q+1)}{2}. \quad (1)$$

Exactly $1 + \mathbf{1}_{2|d}$ of these roots are the degenerate parameters $P = \pm 2$, and exactly $(q-1)/2$ parameters have nonsquare discriminant. The formula is derived self-containedly in Appendix E.1; it is not attributed to Williams and no literature-novelty claim is made.

THM-002 [PROVED]. For every fixed schedule pair satisfying DEF-007, fresh exact uniform parameters $P \bmod K$ give a classical Las Vegas algorithm that completely factors every input in the hereditary nonsplit Lucas-asymmetric promise class, is always correct, terminates almost surely, and runs in expected time polynomial in the binary input length. Each witness trial succeeds with probability at least $1/12$.

At each trial the algorithm first retains a proper factor from

$$\gcd(P^2 - 4, K),$$

then, even if that GCD is all of K , separately retains a proper factor from

$$\gcd(V_d(P, 1) - 2, K).$$

At the witness prime p , $(p-1)/2$ nonsplit parameters force $V_d = 2$. At q , the number of nondegenerate colliding parameters is

$$C_q(d) = \frac{\gcd(d, q-1) + \gcd(d, q+1)}{2} - 2 \leq \frac{3q-9}{4} < \frac{3q}{4}.$$

The CRT event probability is therefore

$$\frac{p-1}{2p} \frac{q - C_q(d)}{q} \geq \frac{1}{3} \frac{q - C_q(d)}{q} > \frac{1}{12}.$$

Fresh parameters give a geometric tail with expected cycle count at most 12. The exponent has $O(B(k) \log B(k) + \log R(k))$ bits, binary Lucas evaluation is polynomial, and the hereditary recursion has $O(m)$ nodes after exact maximal-power preprocessing. Appendix E.2 gives the complete exceptional branch, termination, recursion, and bit-complexity proof.

9 A combined-promise density barrier

Fix one nonempty finite exponent set $\Delta \subset \mathbb{Z}_{>0}$. For the M3 and M7 schedule at input length k , both channels use the same fixed

$$\Delta(k) = \{t \operatorname{lcm}(1, \dots, B(k)) : 1 \leq t \leq R(k)\}.$$

DEF-008 [DEFINITION]. For an odd prime p , define the combined schedule signature

$$\sigma_\Delta(p) = (\mathbf{1}_{p-1|d}, \mathbf{1}_{p+1|d})_{d \in \Delta}.$$

This is an analytical object evaluated at an unknown factor. It is not claimed to be computable from N without factoring. For a finite odd-prime set S , let $H_\Delta(S)$ be the primes with nonzero signature, and write $s = |S|$, $h = |H_\Delta(S)|$, and $D = \max \Delta$.

BAR-003 [PROVED]. For distinct odd primes p, q , the union of the local $p - 1$ and $p + 1$ divisibility-asymmetry promises holds exactly when

$$\sigma_\Delta(p) \neq \sigma_\Delta(q).$$

For a square-free semiprime this local condition is also the hereditary union of DEF-004 and DEF-007. If $s \geq 2$ and all unordered pairs from S use the same Δ , their promised fraction satisfies the finite-distribution upper bound

$$\rho_\Delta(S) \leq \frac{h(2s - h - 1)}{s(s - 1)}, \quad h \leq 2 \sum_{d \in \Delta} \tau(d) \leq 4|\Delta|\sqrt{D}. \quad (2)$$

Thus

$$\rho_\Delta(S) \leq \min \left\{ 1, \frac{8|\Delta|\sqrt{D}}{s - 1} \right\}.$$

Every $p > D + 1$ has zero signature.

For

$$S_n = \{p \text{ odd prime} : 2^n < p < 2^{n+1/2}\},$$

every distinct-prime product has input length $k = 2n + 1$. Consequently, if $|S_n| \geq 2$ and

$$\max \Delta(2n + 1) + 1 < 2^n,$$

the promised-pair density on this explicitly defined balanced-semiprime distribution is exactly zero. More generally the displayed upper bound tends to zero if

$$|\Delta(2n + 1)|\sqrt{\max \Delta(2n + 1)} = o(|S_n|).$$

Appendix F.1 gives the complete elementary proof. No literature-novelty claim is made.

REF-004 [REFUTED]. A common combined schedule cannot cover every balanced semiprime through these asymmetries while its largest exponent remains below each factor minus one. The smallest unrestricted witness is $\Delta = \{1\}$, $N = 15 = 3 \cdot 5$, for which both signatures are zero. This witness is not asserted to lie in the interval defining S_n .

BAR-003 is a magnitude/sparsity barrier, not a recognizer or a lower bound for every polynomial-bit-length schedule. An exponent with $\Theta(k)$ bits may exceed the balanced factors and fall outside its zero-density hypothesis.

10 An exponent-encoding divisor barrier

We now charge the binary representation of explicit exponents rather than their numerical magnitude. For a positive integer d , let $\ell(d) = \lceil \log_2(d + 1) \rceil$. For $\ell \geq 1$, set

$$T_\ell = \lfloor \sqrt{\ell} \rfloor$$

and let A_ℓ be the largest nonnegative integer satisfying

$$(T_\ell + 1)^{A_\ell} < 2^\ell.$$

DEF-009 [DEFINITION]. Define the exact one-length divisor budget and its monotone envelope by

$$B(\ell) = (\ell + 1)^{T_\ell} 2^{A_\ell}, \quad Q(L) = \max_{1 \leq \ell \leq L} B(\ell).$$

Both are integer valued and require no floating-point approximation.

Let $\Delta(k)$ be one factorization-independent explicit list used for every pair in a declared common-input-length population, and write

$$E(k) = |\Delta(k)|, \quad L(k) = \max_{d \in \Delta(k)} \ell(d).$$

Let $H(k)$ be the set of all odd primes having a nonzero combined signature for this schedule.

BAR-004 [PROVED]. Every positive integer d satisfies

$$\tau(d) \leq B(\ell(d)) \leq Q(\ell(d)), \quad \log_2 Q(L) = O\left(\frac{L}{\log L}\right).$$

Consequently,

$$|H(k)| \leq 2 \sum_{d \in \Delta(k)} \tau(d) \leq 2E(k)Q(L(k)).$$

If $E(k) = k^{O(1)}$ and $L(k) = o(k \log k)$, then

$$|H(k)| \leq 2^{o(k)}.$$

In particular, this includes $O(k)$ -bit exponents even when their numerical values exceed the prime factors.

Fix $\alpha > 0$. For any explicitly declared finite odd-prime set S_k whose distinct-prime products all have input length k and whose size satisfies $|S_k| \geq 2^{\alpha k}$, the uniform unordered-pair fraction ρ_k in the local, hence square-free-semiprime hereditary, union of the M3 and M7 promises obeys

$$\rho_k \leq \frac{2|H(k)|}{|S_k| - 1} \leq 2^{-\alpha k + o(k)} \rightarrow 0.$$

Appendix G.1 supplies the complete elementary proof. This is an explicit-list, stipulated-population hit-set bound. It neither constructs such a prime population nor invokes a prime-distribution theorem.

REF-005 [REFUTED]. Numerical value alone does not replace divisor structure. With $\Delta = \{7\}$ and $N = 15 = 3 \cdot 5$, the exponent exceeds both $p + 1 = 4$ and $q + 1 = 6$, but none of 2, 4, 4, 6 divides 7. Both signatures are zero. EXP-0008 confirms that this is the smallest bounded counterexample under the strict value condition.

BAR-004 does not cover exponentially many explicit exponents, adaptive or factor-dependent schedules, compressed implicit or batched representations, or $L(k) \neq o(k \log k)$. It is not a recognizer, a natural-density theorem, a factoring lower bound, or a barrier to other algebraic mechanisms. No novelty or priority claim is made for its elementary divisor estimate.

11 A multiplication straight-line compression barrier

We now specify one compressed evaluation model rather than treating short syntax as automatically cheap.

DEF-010 [DEFINITION]. Fix $N \geq 2$ and $g \in \mathbb{Z}/N\mathbb{Z}$. A factor-oblivious multiplication straight-line program has

$$x_0 = g, \quad x_i = x_{a_i} x_{b_i} \pmod{N} \quad (1 \leq i \leq t),$$

where $0 \leq a_i, b_i < i$. Its formal exponents are

$$e_0 = 1, \quad e_i = e_{a_i} + e_{b_i}.$$

Parent indices, output indices, and every multiplication node are charged. With binary indices the table has $O(t \log(t+1))$ bits; direct evaluation uses exactly t modular multiplications and retains at most $t+1$ residues. The factor-oblivious constructor must emit the tables in polynomial time. This includes repeated squaring and shared addition chains, but excludes inversion, nonmultiplicative operations, special endomorphisms, adaptive factor-dependent branches, and uncharged implicit output expansion.

BAR-005 [PROVED]. Every node satisfies

$$x_i = g^{e_i} \pmod{N}, \quad e_i \leq 2^i.$$

Thus exact formal realization of g^d needs at least $\lceil \log_2 d \rceil$ multiplication nodes. Repeated squaring attains the bound when d is a power of two. A program has at most $t+1$ node outputs, each with exponent bit length at most $t+1$.

For a fixed number of initial bases, let $T(k)$ be the total charged node count of every output-producing program in a common factorization-independent schedule. Then

$$E(k) \leq T(k) + O(1), \quad L(k) \leq T(k) + 1.$$

If $T(k) = o(k \log k)$, BAR-004 applies and gives its subexponential hit-set and stipulated-population vanishing-fraction conclusions. If $T(k) = k^{O(1)}$, the model cannot hide a superpolynomial-bit formal exponent at polynomial multiplication cost.

REF-006 [REFUTED]. Inside DEF-010, the short descriptor

$$\text{tower}(s) = 2^{2^s}$$

does not make exact formal evaluation polynomial in s . BAR-005 requires at least 2^s multiplications, and repeated squaring attains exactly that count.

Appendix H.1 supplies the complete elementary proof. No literature-novelty claim is made; the contribution here is the explicitly scoped application to BAR-004. The result concerns exact formal-exponent realization, not the shortest computation of an equal residue for a fixed N, g . It makes no conclusion for $T(k) \neq o(k \log k)$, richer algebraic representations, promise recognition, natural density, or general factoring.

12 A constant-sensitive boundary barrier

We now retain the leading coefficient at the exact $k \log k$ boundary. For $\ell \geq 1$, put

$$b_\ell = \lceil \log_2(\ell + 1) \rceil, \quad c_\ell = \lceil \log_2(b_\ell + 1) \rceil,$$

$$T_\ell = \max \left\{ 1, \left\lfloor \frac{\ell}{b_\ell^2 c_\ell} \right\rfloor \right\},$$

and let A_ℓ be maximal with $(T_\ell + 1)^{A_\ell} < 2^\ell$.

DEF-011 [DEFINITION]. Define the exact integer budget

$$C(\ell) = (\ell + 1)^{T_\ell} 2^{A_\ell}, \quad R(L) = \max_{1 \leq \ell \leq L} C(\ell).$$

BAR-006 [PROVED]. Every positive d satisfies

$$\tau(d) \leq C(\ell(d)) \leq R(\ell(d)), \quad \log_2 R(L) \leq (1 + o(1)) \frac{L}{\log_2 L}.$$

Consequently, a common factorization-independent schedule with polynomially many explicit exponents and

$$L(k) \leq (c + o(1))k \log_2 k$$

has global combined hit set

$$|H(k)| \leq 2^{ck+o(k)}.$$

On any stipulated common-input-length finite odd-prime population with $|S_k| \geq 2^{\alpha k}$ and fixed $\alpha > c$, BAR-003 gives promised-pair fraction at most

$$2^{-(\alpha-c)k+o(k)} \longrightarrow 0.$$

For a fixed number of bases, the same coefficient applies to DEF-010 schedules whose total charged multiplication-node count has the displayed upper bound. This statement neither constructs S_k nor transfers to natural density.

The boundary is nonvacuous. For

$$P_r = \prod_{j=1}^r p_j,$$

EXT-005 and the elementary lower bound $p_j \geq j + 1$ give $\ell(P_r) = \Theta(r \log r)$, while square-freeness gives $\tau(P_r) = 2^r$. Binary exponentiation uses exactly

$$\ell(P_r) - 1 + \text{popcount}(P_r) - 1 \leq 2\ell(P_r) - 2$$

multiplication nodes, and BAR-005 supplies the matching-order lower bound. Thus P_r is an explicit factor-oblivious boundary family with exponential divisor capacity. Capacity does not prove prime yield: no asymptotic lower bound is known here for divisors $d \mid P_r$ with $d \pm 1$ prime.

Also, P_r contains exactly one factor of two. For every odd prime q , one of $q - 1, q + 1$ is divisible by four, so they cannot both divide P_r . Every nonzero one-exponent signature is $(1, 0)$ or $(0, 1)$, never $(1, 1)$.

REF-007 [REFUTED]. Boundary node count alone is insufficient. Take $t(k) = \lfloor k \log_2 k \rfloor$, repeatedly square, and expose every node exponent $1, 2, 4, \dots, 2^{t(k)}$. Their divisor union contains only $t(k) + 1$ powers of two, hence at most $2(t(k) + 1)$ odd-prime candidates of the form $d \pm 1$. This $\Theta(k \log k)$ -node schedule has a polynomial hit set and therefore a vanishing fraction on every stipulated $2^{\Omega(k)}$ -size population.

Appendix I.1 gives the complete proof. The result is silent when $c \geq \alpha$, supplies no primorial prime-yield theorem or recognizer, and remains inside the explicit-schedule and DEF-010 scopes.

13 Algorithms and bit-complexity synthesis

The mathematical results use distinct algorithmic and analytical scopes that must not be conflated.

Component	Guarantee	Complexity boundary
Explicit multiplicative family	Exact candidate semantics and recursive failure propagation	Polynomial only if construction, cardinalities, exponent lengths, evaluation, and recursive success are all polynomial in $\log N$
Multiplication straight-line family	Exact formal exponents and shared same-base multiplication DAGs	t nodes use $O(t \log(t + 1))$ descriptor bits and t modular multiplications; no fixed-modulus or richer-operation lower bound
Semismooth randomized splitter	Complete Las Vegas factorization on the hereditary promise class	Expected polynomial bit complexity for every fixed polynomial schedule pair B, R ; no outside-promise termination claim
Nonsplit Lucas splitter	Exact discriminant and sequence outcomes; conjugate-map barrier; Las Vegas factorization on the DEF-007 hereditary promise	Binary Lucas evaluation is polynomial in $\log N + \log d$; no outside-promise termination or membership-recognition claim
Reference suite	Exact Python semantics, bounded Rust <code>u64</code> , selected C# <code>BigInteger</code> verification	Validation and falsification only; no practical or asymptotic superiority claim

Binary modular exponentiation uses $O(L(m))$ modular multiplications for an $L(m)$ -bit exponent. A Lucas pair can likewise be evaluated by binary composition in $O(\log d)$ modular operations. Every failed inversion or nonunit precheck is an algorithmic branch that may expose a factor; it is not silently treated as an exceptional implementation error.

For complete factorization, every reported split is verified before recursion. A binary split tree has fewer than $2m$ nodes because the number of prime factors with multiplicity is at most m . In the restricted theorem, maximal-perfect-power preprocessing adds at most one unary node per subsequent split or leaf, preserving an $O(m)$ recursion count.

14 Reproducible experimental methodology

All searches are deterministic exhaustive enumerations inside declared finite boxes; their seed is therefore recorded as none. Prime inputs, nonunit bases, or empty families are skipped only when excluded by the tested definition. The searches retain prime powers, repeated-prime products, Carmichael numbers, orders 1 and 2, simultaneous collisions, discriminant degeneracy, and explicit independently parameterized controls where relevant.

Python arbitrary-precision code is the semantic oracle. Rust independently implements the documented `u64` overlap with `u128` modular products, and C# uses `BigInteger` for selected vectors. A passing differential check establishes agreement only on those registered vectors. The experiment records name the core commit, exact command, bounds, pruning, toolchain manifest, and canonical summary hash.

The baseline suite includes modular exponentiation, trial division, perfect-power detection, validation primality, bounded Pollard rho, Pollard $p - 1$, a scoped Williams-style $p + 1$ stage, and per-item batch-GCD semantics. The batch routine is not presented as a product-tree optimization.

15 Results

Record	Finite scope	Registered observation
Baseline	Canonical arithmetic and factoring vectors	58 cross-language checks passed
EXP-0002	$N \leq 500, g \leq 20, d \leq 20$	78,860 unit-base candidates; 5,672 nonsquarefree support-only false negatives; 24 differential checks
EXP-0003	$N \leq 500, B = 8, R = 3$	155 hereditary promised inputs factored by the finite oracle; 557 witnesses met the 5/12 bound; 22 differential checks
EXP-0004	4,095 subsets of [12], orders through 8	114,660 two-order pair-profile checks; 576 covers, including 240 noninjective covers; 12 differential checks
EXP-0005	$N \leq 700$, bases and parameters through 32, $d \leq 12$	117,276 identities; zero derived-Lucas-only family gains; 18 differential checks
EXP-0006	13 odd primes through 43, $d \leq 80$	1,040 root formulas and 75,934 proved-event splits; minimum event probability 8/51; 26 differential checks
EXP-0007	987 exponent families; 25 odd primes through 101	296,100 direct/signature comparisons; 184,994 magnitude-zero pair checks; 28 differential checks
EXP-0008	Every $1 \leq d < 2^{18}$; direct primes through 4093	262,143 divisor budgets; 2,306,048 direct hit-oracle checks; 987 record-family bounds; 46 differential checks
EXP-0009	All 1,587,600 seven-node programs; tower levels through 16	1,647,202 exponent-growth and residue checks; 17 tower checks; 24 differential checks
EXP-0010	Every $1 \leq d < 2^{18}$; primorials through $r = 12$	262,143 boundary-budget checks; 8,157 distinct prime candidates; 32 differential checks

EMP-001 [EMPIRICAL]. The baseline corpus passed 58 registered cross-language comparisons. This is finite implementation agreement.

EMP-002 [EMPIRICAL]. In EXP-0002, all 78,860 candidates agreed with the support-sufficiency and exact valuation criteria. The search found 5,672 nonsquarefree proper GCDs not described by a proper support, beginning with $(N, g, d) = (4, 3, 1)$.

EMP-003 [EMPIRICAL]. In EXP-0003, the finite oracle completely factored all 155 hereditary promised inputs in the box, and all 557 ordered witnesses met the proved 5/12 success bound. The minimum observed exact probability was 268/493. This does not estimate promise-class density.

EMP-004 [EMPIRICAL]. In EXP-0004, 240 of 576 divisor covers had noninjective signatures. The exact signature characterization, counting bound, and control construction passed their registered checks.

EMP-005 [EMPIRICAL]. In EXP-0005, the multiplicative and combined families both succeeded on exactly 9,037 of 9,773 conjugate families, with zero derived-Lucas-only gains. Independently chosen parameters produced bounded complement examples, but the tuple counts are not probabilities.

EMP-006 [EMPIRICAL]. In EXP-0006, all 1,040 exact root formulas, 714 ordered asymmetry witnesses, and 75,934 proved-event splitting parameters passed. The minimum proved-event probability was $8/51$, and 26 selected Python/Rust/C# comparisons agreed. These finite counts do not estimate promise-class density or establish novelty.

EMP-007 [EMPIRICAL]. In EXP-0007, all 296,100 direct/signature comparisons, 987 density and divisor-bound checks, 184,994 magnitude-zero pair checks, and 2,443 balanced zero-density checks passed. The maximum observed promised-pair fraction was $49/100$, and 28 selected Python/Rust/C# comparisons agreed. These registered finite-distribution observations are not a natural-density estimate, recognizer, asymptotic result, or novelty claim.

EMP-008 [EMPIRICAL]. In EXP-0008, all 262,143 exact divisor enumerations, one-length budgets, and single-exponent hit bounds passed, as did 2,306,048 direct prime-oracle comparisons and 987 record-family bounds. The largest observed single-exponent hit set had 114 odd primes at $d = 166,320$, and 46 selected Python/Rust/C# comparisons agreed. These finite observations do not establish budget tightness, asymptotic behavior, compressed-representation behavior, a natural density, or novelty.

EMP-009 [EMPIRICAL]. In EXP-0009, all 1,647,202 constructed nodes satisfied the exponent-growth bound and direct residue semantics. Every depth through seven attained 2^t ; all 17 tower levels through $s = 16$ matched the exact lower bound and repeated-squaring construction; and 24 selected Python/Rust/C# comparisons agreed. These finite syntax and residue checks do not establish an asymptotic, richer-model, density, novelty, or factoring claim.

EMP-010 [EMPIRICAL]. In EXP-0010, all 262,143 exact DEF-011 budget checks passed. Every divisor and $d \pm 1$ candidate was enumerated for the first 12 primorials. At $r = 12$, the 43-bit exponent had 4,096 divisors, 67 binary multiplication nodes, and 897 prime hits split $449/448$ between the two disjoint channels. The within-hit promised fraction was $449/897$, and 32 selected Python/Rust/C# comparisons agreed. These finite records do not establish asymptotic prime yield or a fraction on an external population.

These observations audit implementations and minimize counterexamples. The universal statements in this paper are supplied by proofs, not extrapolation from the table.

16 Limitations and open problems

No universal support-POSF or valuation-family constructor is supplied. THM-001 applies only to a factor-dependent hereditary promise and provides neither a membership recognizer nor a density theorem. Its unbounded splitter may not terminate outside the promise.

THM-002 has the same promise-versus-recognition boundary for a different, odd $p + 1$ -asymmetry class. It requires a fresh exact uniform parameter at every attempt, supplies no outside-promise termination or density guarantee, and does not assert statistical independence from Pollard $p - 1$.

BAR-001 is a theorem about explicit divisibility signatures, not a lower bound for every compressed batch representation or for the Umans–Wang collision-resolution mechanism. Its order profile is factor dependent and is used for analysis, not as an algorithmic input.

BAR-002 concerns only $P = a + a^{-1}$. THM-002 shows that independent uniform Lucas parameters do have a quantified success probability under DEF-007, but it does not provide a universal

construction. Neither result rules out quadratic norm-one variants, elliptic-curve channels, adaptive selection, or another parameter map.

BAR-003 concerns one common explicit finite exponent schedule and one uniform distribution on unordered pairs from a declared finite prime set. Its signatures require the unknown factors and therefore do not recognize promise membership. The exact zero-density result requires its strict magnitude hypothesis, while the asymptotic upper bound requires the displayed sparsity hypothesis. It is neither a natural-density theorem nor a lower bound for schedules whose polynomial-bit-length exponents have larger values.

BAR-004 extends the obstruction to polynomial-size explicit lists with $L(k) = o(k \log k)$, including $O(k)$ -bit exponent values above the factors. Its vanishing fraction is conditional on a stipulated exponentially large finite prime population at common input length; the paper neither constructs that population nor transfers the statement to natural semiprime density. The result charges individual exponent encodings and evaluations. It does not cover compressed or batched implicit families, adaptive factor-dependent choices, $L(k) \neq o(k \log k)$, exponentially many explicit exponents, or other algebraic mechanisms, and is not a factoring lower bound.

BAR-005 closes one of those compressed cases only. It charges every node in a factor-oblivious multiplication-only same-base DAG and tracks exact formal exponents. It is not a lower bound for computing an equal residue at one fixed modulus and base, for inversion or addition-subtraction chains, for special endomorphisms, for adaptive factor-revealing branches, or for another explicitly costed algebraic model.

BAR-006 resolves only a constant-sensitive part of the $\Theta(k \log k)$ -node boundary. For a stipulated population exponent α , it proves vanishing below coefficient α , but says nothing at or above that coefficient. The primordial family proves exponential divisor capacity, not an asymptotic lower bound on actual $d \pm 1$ prime hits. REF-007 shows that boundary node count alone is not sufficient. None of these statements constructs the population, recognizes the promise, or supplies a factoring lower bound.

The authoritative Rust layer is bounded to `u64`; validation primality uses trial division and is not itself the polynomial-time procedure imported in EXT-001. The literature set is sufficient for the technical statements used here but is not a comprehensive current-state-of-the-art survey.

The main open problems are the two explicit universal constructor questions in OPEN-002 and OPEN-003, recognition of the combined hereditary promise without access to the factorization, whether any explicit boundary family has a proved $d \pm 1$ prime-yield guarantee strong enough at or above the critical coefficient, and whether a richer explicitly costed representation can escape BAR-004–BAR-006.

17 Conclusion

Order-separating exponent research must distinguish hitting an order from separating prime-power components. Fresh randomized bases turn a factor-dependent $p - 1$ asymmetry into an expected-polynomial Las Vegas theorem on one hereditary promise class. An exact Lucas root count turns fresh uniform parameters and a $p + 1$ asymmetry into a second restricted theorem. Divisor coverage alone still fails because signatures can collide, and the natural conjugate Lucas map still fails as an independent channel because it algebraically squares the multiplicative residue. A common $p - 1 / p + 1$ schedule also covers no balanced-semiprime pair once every factor exceeds its largest exponent by more than one, and its promised fraction is bounded by the schedule’s divisor hit set. Charging exponent encodings sharpens this: a polynomial-size explicit list with $o(k \log k)$ -bit members has a subexponential global hit set, so its promised fraction vanishes on every stipulated exponentially

large common-input-length prime population. Merely making an exponent numerically larger than the factors does not help unless the required $p \pm 1$ divisor structure is present. Nor does short symbolic syntax alone help in DEF-010: formal exponent bit length grows by at most one per charged multiplication, so the tower descriptor 2^{2^s} still costs exactly 2^s multiplications. At the exact boundary, the optimized divisor budget transfers the exponent-length coefficient c into a $2^{ck+o(k)}$ hit-set bound. This still vanishes below a stipulated population coefficient α . Primorials show that exponential divisor capacity exists at boundary cost, but neither that capacity nor the node count alone supplies a prime-yield guarantee.

These results do not resolve general classical factoring. They replace plausible shortcuts with exact obligations: recognizable or explicitly promised input structure, injective rather than merely nonempty signatures, genuinely nonsplit rather than conjugately re-encoded group parameters, and divisor-rich schedules whose representation, construction, and evaluation costs are all accounted for. The exact $\Theta(k \log k)$ straight-line boundary now has a subcritical coefficient obstruction and an explicit capacity witness; the critical and supercritical prime-yield problem remains a research target, not a claimed factoring breakthrough.

A Full proofs of the separator criteria

A.1 Proof of LEM-001

Choose $p \in D_N(g, d)$. By the definition of multiplicative order, $g^d \equiv 1 \pmod{p}$, so $p \mid \gcd(g^d - 1, N)$ and the GCD exceeds one. Because the support is proper, choose $q \in P(N) \setminus D_N(g, d)$. Then $g^d \not\equiv 1 \pmod{q}$, so q does not divide the GCD. Since $q \mid N$, the GCD is strictly smaller than N . Repeated prime factors affect neither strict inequality. $\square$

A.2 Proof of LEM-002

For each prime p , the exponent of p in a GCD is the minimum of its exponents in the two arguments. Its exponent in N is e_p , while its exponent in $g^d - 1$ is $\nu_p(g^d - 1)$, including the convention for zero. Unique factorization gives

$$\gcd(g^d - 1, N) = \prod_{p \in P(N)} p^{\min(e_p, \nu_p(g^d - 1))}.$$

The product is one exactly when every capped exponent is zero, and it is N exactly when every capped exponent is e_p . These are the only trivial GCDs.

If N is square-free, every $e_p = 1$, and

$$a_p = 1 \iff p \mid g^d - 1 \iff \text{ord}_p(g) \mid d.$$

Thus support and valuation separation coincide. If $N = p^e$, the support can never be proper, while the valuation criterion is precisely $0 < \nu_p(g^d - 1) < e$. $\square$

B Full proof of the restricted theorem

B.1 Proof of THM-001

Algorithm. For a reached composite non-perfect-power K , put $k = \lceil \log_2 K \rceil$ and $M = M_B(k)$. Repeat complete cycles over $t = 1, \dots, R(k)$. At every t , sample a fresh exact uniform $a \in$

$\{0, \dots, K-1\}$, return $\gcd(a, K)$ if it is proper, and otherwise return $\gcd(a^{tM} - 1, K)$ if that GCD is proper.

The complete factorer first runs deterministic polynomial-time primality testing. It next performs exact maximal-exponent perfect-power detection. If $K = b^e$, it recursively factors b and multiplies every returned multiplicity by e . At every other composite, it invokes the randomized splitter, validates the returned divisor, and recursively factors both children.

Exact uniform sampling uses rejection: draw k unbiased bits as $x \in [0, 2^k - 1]$ and reject $x \geq K$. Since $2^{k-1} < K \leq 2^k$, acceptance probability exceeds $1/2$, so fewer than two k -bit draws are needed in expectation.

One-trial success. Fix a promise witness (p, q, t) and let $d = tM_B(k)$. If $1 < \gcd(a, K) < K$, the first GCD succeeds. The only residue whose GCD with K is K is $a = 0$.

For a unit a , Fermat's theorem and $p-1 \mid d$ give $a^d \equiv 1 \pmod{p}$. Projection of a uniform unit modulo K to $(\mathbb{Z}/q\mathbb{Z})^\times$ is uniform. This cyclic group has order $q-1$, so $x^d = 1$ has exactly $\gcd(d, q-1)$ roots. Because $q-1 \nmid d$, that proper divisor of $q-1$ is at most $(q-1)/2$. At least half the units therefore satisfy $a^d \not\equiv 1 \pmod{q}$, and each such unit separates p from q .

There are $K - \varphi(K) - 1$ proper nonunit residues and at least $\varphi(K)/2$ successful units. Hence

$$\Pr[\text{success}] \geq \frac{K - 1 - \varphi(K)/2}{K} \geq \frac{K - 1}{2K}.$$

A composite non-perfect-power has at least two distinct prime divisors, so $K \geq 6$, and the final quantity is at least $5/12$.

Correctness and termination. Every returned split is explicitly checked to lie strictly between 1 and the current input. Prime leaves are recognized deterministically, and exact perfect-power expansion preserves multiplicities. The algorithm therefore never returns an incorrect factorization.

One value of t in each cycle is a witness. Conditional on any prior history, its fresh sample succeeds with probability at least $5/12$. Consequently, the probability of surviving s full cycles is at most $(7/12)^s$, which tends to zero, and the expected number of cycles is at most $12/5$. Every child divides N , so the hereditary promise applies at every recursive node. Induction on the input value gives complete correctness, and the geometric tail gives almost-sure termination.

Expected bit complexity. Since $M_B(k) \mid B(k)!$,

$$\log_2 M_B(k) \leq \log_2(B(k)!) \leq B(k) \log_2 B(k).$$

Iterative LCM construction uses $B(k) - 1$ GCD/multiplication steps, and every intermediate value divides $B(k)!$. Each $d = tM_B(k)$ therefore has

$$O(B(k) \log B(k) + \log R(k))$$

bits. One full cycle performs $R(k)$ exact samples, modular exponentiations, and GCD computations. The schedule functions and all these operations cost polynomially many bit operations by hypothesis. The expected cycle count is constant.

Exact maximal-power detection is polynomial: enumerate $2 \leq e \leq k$, find the integer e -th root by binary search with products capped above K , and verify equality. Choosing the maximal exponent leaves a base that is not a perfect power, so unary perfect-power nodes cannot be consecutive. Charge each unary node to the following split node or prime leaf. The binary split tree has fewer than $2m$ nodes, and the full recursion fewer than $4m$ invocations. Monotonicity bounds every local schedule by its value at m ; linearity of expectation yields polynomial expected total time and polynomial space. $\square$

C Full proof of the coverage-separation barrier

C.1 Proof of BAR-001

For a profile $R = (r_1, \dots, r_s)$, suppose d separates the profile. There is a hit index i and a miss index j , so $d \in \Sigma_\Delta(r_i)$ and $d \notin \Sigma_\Delta(r_j)$. The signatures are not all equal. Conversely, if two signatures differ, an element of their symmetric difference is divisible by one associated order but not the other. Its support on the full profile is nonempty and proper. Thus a profile is separated exactly when its signatures are not all equal.

It follows that every profile containing two distinct values from $\{1, \dots, n\}$ is separated exactly when the signature map is injective there. The n -divisor property requires only that every signature be nonempty, so it cannot imply injectivity. The explicit counterexample $\Delta = \{2\}$ has

$$\Sigma_\Delta(1) = \Sigma_\Delta(2) = \{2\}.$$

It is realized as the positive difference of $S = \{3\}$ and $T = \{1\}$, and the $N = 6, g = 5$ calculation in the main text realizes the simultaneous multiplicative collision.

If $k = |\Delta|$, there are only $2^k - 1$ nonempty subsets of Δ . Coverage plus injectivity on n orders requires n distinct nonempty signatures, so

$$n \leq 2^k - 1, \quad k \geq \lceil \log_2(n + 1) \rceil.$$

Every positive difference is generated by a pair in $S \times T$, hence $|\Delta| \leq |S||T|$, proving the second bound.

For the control construction, put $h = \lceil \sqrt{n} \rceil$ and

$$S = \{1 + ih : 1 \leq i \leq h\}, \quad T = \{1 + j : 0 \leq j < h\}.$$

Its positive differences are

$$\{ih - j : 1 \leq i \leq h, 0 \leq j < h\} = \{1, \dots, h^2\}.$$

For distinct $r < s \leq n$, candidate $d = r$ is divisible by r but not by s , so it separates. The construction uses $|S||T| = h^2 = \Theta(n)$ pairs. It proves sufficiency but is exponential when n is of magnitude exponential in the binary input length. $\square$

D Full proof of the conjugate-channel barrier

D.1 Proof of BAR-002

Put $W_d = a^d + a^{-d}$ in $\mathbb{Z}/N\mathbb{Z}$. Then

$$W_0 = 2, \quad W_1 = a + a^{-1} = P(a),$$

and direct multiplication gives

$$W_{d+1} = P(a)W_d - W_{d-1}.$$

The initial values and recurrence uniquely determine the Lucas sequence, so

$$V_d(P(a), 1) = a^d + a^{-d} \pmod{N}.$$

Expanding gives

$$V_d(P(a), 1) - 2 = a^{-d}(a^d - 1)^2 \pmod{N}$$

and

$$P(a)^2 - 4 = (a - a^{-1})^2 = a^{-2}(a^2 - 1)^2 \pmod{N}.$$

Let $p^e \parallel N$. Because a is a unit, the inverse factors have p -adic valuation zero. Congruence modulo p^e therefore gives the capped identity

$$\min\{e, \nu_p(V_d(P(a), 1) - 2)\} = \min\{e, 2\nu_p(a^d - 1)\}.$$

Doubling preserves whether the capped valuation is positive, so the sequence residues have identical prime support. If N is square-free, every cap is one, so the capped valuations, and hence the two raw GCDs, are equal.

Suppose $G_L(N, P(a), d)$ is proper. Its capped profile is neither all zero nor full. The undoubled multiplicative profile is positive at exactly the same primes; wherever the doubled profile is not full, the undoubled profile is also not full. Thus $G_M(N, a, d)$ is proper. If $G_\Delta(N, P(a))$ is proper, the discriminant identity applies the same argument to $a^2 - 1$, so $G_M(N, a, 2)$ is proper. When 2 belongs to the exponent family, that candidate is present.

The combined family retains every multiplicative candidate. It succeeds whenever the multiplicative family succeeds, and the preceding implications show that it cannot succeed on an additional input. Their success domains are equal. $\square$

E Full proofs of the nonsplit Lucas results

E.1 Proof of LEM-003

First recall the finite-field structure used in the count. Every finite multiplicative subgroup H of a field is cyclic. If e is its exponent, then all $|H|$ elements are roots of $X^e - 1$, so $|H| \leq e$. Combining elements of maximal prime-power order in the finite abelian group gives an element of order e , while $e \leq |H|$; hence $e = |H|$. Thus $\mathbb{F}_q^\times$ and $\mathbb{F}_{q^2}^\times$ are cyclic of orders $q - 1$ and $q^2 - 1$. The norm map $x \mapsto x^{q+1}$ sends a generator of the latter to an element of order $q - 1$, so its kernel is cyclic of order $q + 1$.

Let α be a root of $X^2 - PX + 1$. The other root is α^{-1} , and the recurrence gives

$$V_d(P, 1) = \alpha^d + \alpha^{-d}.$$

For a nonzero square discriminant, $\alpha \in \mathbb{F}_q^\times$; for a nonsquare discriminant, α belongs to the norm-one subgroup. In either group,

$$\alpha^d + \alpha^{-d} = 2 \iff (\alpha^d - 1)^2 = 0 \iff \alpha^d = 1.$$

A cyclic group of order r has $\gcd(d, r)$ such roots. Away from $\alpha = \pm 1$, the map $\alpha \mapsto P = \alpha + \alpha^{-1}$ has two-element fibers $\{\alpha, \alpha^{-1}\}$. The element 1 contributes $P = 2$; the element -1 contributes $P = -2$ exactly for even d . Pairing all other roots and counting these shared degenerate parameters once gives

$$\#\{P : V_d(P, 1) = 2\} = \frac{\gcd(d, q-1) + \gcd(d, q+1)}{2}$$

and the stated degenerate-root count.

It remains to count nonsplit parameters. The invertible substitution $u = P - y$, $v = P + y$ identifies the solutions of $y^2 = P^2 - 4$ with the $q - 1$ pairs $uv = 4$. Therefore

$$\sum_{P \in \mathbb{F}_q} \left(1 + \left(\frac{P^2 - 4}{q} \right) \right) = q - 1,$$

so the character sum is -1 . There are two zero-discriminant parameters. If n_+ and n_- count the nonzero square and nonsquare cases, then

$$n_+ + n_- = q - 2, \quad n_+ - n_- = -1,$$

and hence $n_- = (q - 1)/2$. $\square$

E.2 Proof of THM-002

Algorithm. On a reached integer K , first run deterministic polynomial-time primality testing, exact maximal-perfect-power preprocessing, and the even factor check. For an odd composite non-perfect-power residual, put $k = \lceil \log_2 K \rceil$, $M = M_B(k)$, and repeat complete cycles over $t = 1, \dots, R(k)$. At every t , sample a fresh exact uniform $P \in \{0, \dots, K - 1\}$, put $d = tM$, and compute

$$h_\Delta = \gcd(P^2 - 4, K).$$

Return it if proper. In every other branch, including $h_\Delta = K$, compute

$$h_L = \gcd(V_d(P, 1) - 2, K)$$

and return it if proper. Validate every split and recurse on both children. Uniform rejection sampling accepts a k -bit draw with probability greater than $1/2$.

One-trial success. Fix a witness (p, q, t) . Projection of a uniform residue modulo K to $\mathbb{F}_p \times \mathbb{F}_q$ is uniform, including when K has repeated prime factors or additional factors. At p , LEM-003 supplies $(p - 1)/2$ nonsquare-discriminant parameters. Their roots lie in the norm-one group of order $p + 1$, so $p + 1 \mid d$ gives

$$V_d(P, 1) = 2 \pmod{p}.$$

Since d is even, the number of nondegenerate parameters modulo q that also satisfy $V_d = 2$ is

$$C_q(d) = \frac{\gcd(d, q - 1) + \gcd(d, q + 1)}{2} - 2.$$

The first GCD is at most $q - 1$. Because $q + 1 \nmid d$, the second is a proper divisor of $q + 1$, hence at most $(q + 1)/2$. Thus

$$C_q(d) \leq \frac{3q - 9}{4} < \frac{3q}{4}.$$

Call every other q -parameter safe. If it is discriminant-degenerate, the discriminant GCD contains q but omits the nonsplit prime p , and is proper. If it is nondegenerate, any discriminant factor is again proper because it omits p ; otherwise the sequence GCD contains p and omits q . Every parameter in the event therefore splits K , and its probability is

$$\frac{p - 1}{2p} \frac{q - C_q(d)}{q} \geq \frac{1}{3} \frac{q - C_q(d)}{q} > \frac{1}{12}.$$

Correctness and termination. Only divisors strictly between 1 and the current input are returned. Primality and perfect-power preprocessing preserve factor multiplicities. At least one t per cycle is a witness, and its parameter is fresh conditional on every prior outcome. The probability of surviving s cycles is at most $(11/12)^s$, so termination is almost sure and the expected cycle count is at most 12. The hereditary promise applies at every reached odd composite non-perfect-power child.

Expected bit complexity. Since $M_B(k) \mid B(k)!$,

$$\log_2 M_B(k) = O(B(k) \log B(k)),$$

and every exponent has $O(B(k) \log B(k) + \log R(k))$ bits. LCM construction, exact sampling, binary Lucas evaluation, and GCD computation are polynomial for the fixed polynomial schedule. One cycle has $R(k)$ trials and the expected number of cycles is constant. Exact maximal-power preprocessing is polynomial by bounded integer-root search. Unary perfect-power nodes can be charged to the following split or leaf, while the binary split tree has $O(m)$ nodes. Linearity of expectation gives polynomial expected total time and polynomial space. $\square$

F Full proof of the combined-promise density barrier

F.1 Proof of BAR-003

For distinct odd primes p, q , a local $p - 1$ or $p + 1$ asymmetry exists exactly when some coordinate of $\sigma_\Delta(p)$ is one and the corresponding coordinate of $\sigma_\Delta(q)$ is zero, in either orientation. This is equivalent to $\sigma_\Delta(p) \neq \sigma_\Delta(q)$. A square-free semiprime has no other composite non-perfect-power divisor, so the local and hereditary-union conditions agree.

Now let S contain $s \geq 2$ odd primes, of which h have nonzero signature. Every pair wholly outside the hit set has two zero signatures and is not promised. Therefore the number of promised pairs is at most

$$\binom{s}{2} - \binom{s-h}{2},$$

which, after division by $\binom{s}{2}$, gives

$$\rho_\Delta(S) \leq \frac{h(2s - h - 1)}{s(s - 1)}. \quad (3)$$

This need not be equality because two nonzero signatures may coincide.

If p lies in the hit set, then $p - 1 \mid d$ or $p + 1 \mid d$ for some $d \in \Delta$. For fixed d , every first-type candidate has the form $r + 1$ and every second-type candidate the form $r - 1$, where $r \mid d$. Before imposing primality or membership in S , there are at most $2\tau(d)$ candidates. Hence

$$h \leq 2 \sum_{d \in \Delta} \tau(d).$$

Pairing divisors below and above $\sqrt{d}$ gives $\tau(d) \leq 2\sqrt{d}$, so, with $D = \max \Delta$,

$$h \leq 4|\Delta|\sqrt{D}.$$

The right side of (3) is at most $2h/(s - 1)$; substitution and the trivial probability bound one prove the stated sparse hit-set bound.

If $p > D + 1$, then both $p - 1$ and $p + 1$ exceed every $d \in \Delta$, so neither divides any scheduled exponent and the signature is zero. Thus a prime set wholly above $D + 1$ has promised-pair density zero.

Finally, for distinct $p, q \in S_n$,

$$2^{2n} < pq < 2^{2n+1}.$$

Because an odd semiprime is not a power of two, its binary input length is $\lceil \log_2(pq) \rceil = 2n + 1$, and all pairs use the same schedule. The strict magnitude hypothesis puts every prime above $D + 1$, giving exact zero density. The conditional asymptotic statement follows directly from the sparse hit-set bound. $\square$

G Full proof of the exponent-encoding divisor barrier

G.1 Proof of BAR-004

Write

$$d = \prod_i p_i^{a_i}, \quad \ell = \ell(d), \quad T = T_\ell.$$

Since $d < 2^\ell$, every $a_i \leq \ell$. There are at most T prime divisors $p_i \leq T$, and each contributes at most $\ell + 1$ to $\tau(d) = \prod_i (a_i + 1)$. Their total contribution is at most $(\ell + 1)^T$.

For the remaining primes put $A = \sum_{p_i > T} a_i$. Each such prime is at least $T + 1$, so

$$(T + 1)^A \leq \prod_{p_i > T} p_i^{a_i} \leq d < 2^\ell.$$

Thus $A \leq A_\ell$. Since $a + 1 \leq 2^a$ for every nonnegative integer a , the large-prime contribution is at most $2^A \leq 2^{A_\ell}$. Therefore

$$\tau(d) \leq B(\ell(d)) \leq Q(\ell(d)).$$

As $\ell \rightarrow \infty$, $T_\ell = O(\sqrt{\ell})$, while the defining strict inequality gives

$$A_\ell < \frac{\ell}{\log_2(T_\ell + 1)} = O\left(\frac{\ell}{\log \ell}\right).$$

Consequently,

$$\log_2 B(\ell) = T_\ell \log_2(\ell + 1) + A_\ell = O\left(\frac{\ell}{\log \ell}\right).$$

The function $x/\log x$ is eventually increasing. Enlarging the constant for finitely many small arguments and taking the maximum over $1 \leq \ell \leq L$ gives

$$\log_2 Q(L) = O\left(\frac{L}{\log L}\right).$$

If an odd prime p has nonzero signature for $\Delta(k)$, then some $d \in \Delta(k)$ satisfies $p - 1 \mid d$ or $p + 1 \mid d$. For fixed d , each candidate prime is one more or one less than a positive divisor of d , so there are at most $2\tau(d)$ candidates before primality is imposed. Hence

$$|H(k)| \leq 2 \sum_{d \in \Delta(k)} \tau(d) \leq 2E(k)Q(L(k)).$$

Suppose $E(k) = k^{O(1)}$ and $L(k) = o(k \log k)$. To avoid any monotonicity assumption on L , choose a fixed L_0 beyond the increasing range of $x/\log x$. Pointwise for sufficiently large k , if $L(k) \leq L_0$, then $Q(L(k))$ is bounded. If $L_0 < L(k) \leq k$, then $\log_2 Q(L(k)) = O(k/\log k) = o(k)$. If $L(k) > k$, then

$$\frac{L(k)}{\log L(k)} \leq \frac{L(k)}{\log k} = o(k).$$

Thus in all cases $\log_2 Q(L(k)) = o(k)$. The polynomial factor $E(k)$ is absorbed into the exponent, proving $|H(k)| \leq 2^{o(k)}$.

Finally let $h_k = |H(k) \cap S_k|$. Every promised pair from S_k intersects this set by BAR-003, so

$$\rho_k \leq \frac{\binom{|S_k|}{2} - \binom{|S_k| - h_k}{2}}{\binom{|S_k|}{2}} \leq \frac{2|H(k)|}{|S_k| - 1}.$$

The assumptions $|S_k| \geq 2^{\alpha k}$ and $|H(k)| \leq 2^{o(k)}$ give

$$\rho_k \leq 2^{-\alpha k + o(k)} \longrightarrow 0.$$

No existence or distribution assertion about the stipulated sets S_k is used. $\square$

H Full proof of the multiplication straight-line compression barrier

H.1 Proof of BAR-005

The residue identity follows by induction. At node zero it is the definition. If it holds for all earlier nodes, then

$$x_i = x_{a_i} x_{b_i} = g^{e_{a_i}} g^{e_{b_i}} = g^{e_{a_i} + e_{b_i}} = g^{e_i} \pmod{N}.$$

For exponent growth, $e_0 = 1 = 2^0$. Both parents of node i have indices at most $i - 1$, so the induction hypothesis gives

$$e_i = e_{a_i} + e_{b_i} \leq 2^{a_i} + 2^{b_i} \leq 2^{i-1} + 2^{i-1} = 2^i.$$

Thus a t -multiplication program that realizes formal exponent d has $d \leq 2^t$, hence

$$t \geq \lceil \log_2 d \rceil \geq \ell(d) - 1.$$

Repeated squaring chooses $(a_i, b_i) = (i - 1, i - 1)$ and attains $e_i = 2^i$, proving tightness for powers of two.

Only the $t + 1$ existing nodes are available as outputs. Summing charged nodes across a fixed number of initial-base programs gives

$$E(k) \leq T(k) + O(1), \quad L(k) \leq T(k) + 1.$$

When $T(k) = o(k \log k)$, the first quantity is polynomially bounded and the second is $o(k \log k)$, so substitution into BAR-004 proves the stated hit-set and stipulated-population consequences. When $T(k) = k^{O(1)}$, the displayed bit-length bound excludes a superpolynomial-bit formal exponent.

Finally,

$$\log_2 \text{tower}(s) = \log_2(2^{2^s}) = 2^s.$$

The lower bound therefore requires 2^s multiplications, while that many repeated squarings attain the target. This proves REF-006 inside DEF-010. $\square$

I Full proof of the constant-sensitive boundary barrier

I.1 Proof of BAR-006

Write

$$d = \prod_i p_i^{a_i}, \quad \ell = \ell(d), \quad T = T_\ell.$$

There are at most T prime factors $p_i \leq T$. Since $2^{a_i} \leq d < 2^\ell$, each small-prime contribution $a_i + 1$ is at most $\ell + 1$, so their product is at most $(\ell + 1)^T$.

For the remaining primes put $A = \sum_{p_i > T} a_i$. Every such prime is at least $T + 1$, whence

$$(T + 1)^A \leq d < 2^\ell.$$

Thus $A \leq A_\ell$. Since $a + 1 \leq 2^a$ for every positive integer a , the large-prime contribution to $\tau(d)$ is at most $2^A \leq 2^{A_\ell}$. Therefore

$$\tau(d) \leq C(\ell(d)) \leq R(\ell(d)). \quad (11)$$

As $\ell \rightarrow \infty$,

$$b_\ell = \log_2 \ell + O(1), \quad c_\ell = \log_2 \log_2 \ell + O(1),$$

and

$$T_\ell = \frac{\ell}{b_\ell^2 c_\ell} (1 + o(1)).$$

It follows that

$$T_\ell \log_2(\ell + 1) = o\left(\frac{\ell}{\log_2 \ell}\right). \quad (12)$$

The strict inequality defining A_ℓ gives

$$A_\ell < \frac{\ell}{\log_2(T_\ell + 1)} \leq (1 + o(1)) \frac{\ell}{\log_2 \ell}, \quad (13)$$

because

$$\log_2(T_\ell + 1) = \log_2 \ell - O(\log_2 \log_2 \ell).$$

Equations (12)–(13) prove the one-length leading bound.

It remains to justify the maximum in R . The exact inequalities $T_\ell b_\ell \leq \ell$ and $A_\ell < \ell$ give

$$\log_2 C(\ell) \leq 2\ell. \quad (14)$$

For $\ell \leq L/(\log_2 L)^2$, equation (14) is $o(L/\log_2 L)$. For larger ℓ ,

$$\log_2 \ell = \log_2 L - O(\log_2 \log_2 L),$$

so the one-length estimate is uniform over this range, and the eventual monotonicity of $x/\log_2 x$ gives

$$\log_2 R(L) \leq (1 + o(1)) \frac{L}{\log_2 L}. \quad (15)$$

Every prime in the global hit set is one more or one less than a positive divisor of some scheduled exponent. Hence

$$|H(k)| \leq 2E(k)R(L(k)).$$

If $L(k) \leq (c + o(1))k \log_2 k$, monotonicity lets us substitute the displayed upper envelope into (15), obtaining

$$\log_2 R(L(k)) \leq ck + o(k).$$

For $c = 0$, use the BAR-004 case split instead: bounded $L(k)$ is trivial; when $L(k) \leq k$, equation (15) gives $O(k/\log k) = o(k)$, and when $L(k) > k$,

$$\frac{L(k)}{\log_2 L(k)} \leq \frac{L(k)}{\log_2 k} = o(k).$$

The polynomial $E(k)$ contributes only $O(\log k) = o(k)$, proving the hit-set bound.

Let $h_k = |H(k) \cap S_k|$. BAR-003 gives

$$\rho_k \leq \frac{\binom{|S_k|}{2} - \binom{|S_k| - h_k}{2}}{\binom{|S_k|}{2}} \leq \frac{2|H(k)|}{|S_k| - 1}.$$

Under $|S_k| \geq 2^{\alpha k}$ with $\alpha > c$, this is

$$\rho_k \leq 2^{-(\alpha - c)k + o(k)} \longrightarrow 0.$$

BAR-005 supplies $E(k) \leq T(k) + O(1)$ and $L(k) \leq T(k) + 1$ for the stated fixed-base DEF-010 schedule, proving its consequence.

For completeness, let $P_r = \prod_{j \leq r} p_j$. EXT-005 gives

$$\log_2 P_r \leq r \log_2 p_r = O(r \log r).$$

Since $p_j \geq j + 1$, the final half of the factors give the matching $\Omega(r \log r)$ lower bound. Square-freeness gives $\tau(P_r) = 2^r$. Scanning its remaining binary digits performs $\ell(P_r) - 1$ squarings and $\text{popcount}(P_r) - 1$ multiplications by the base, proving the displayed upper bound; BAR-005 gives the matching order. Finally, if both $q - 1$ and $q + 1$ divided P_r , then $4 \mid P_r$, because one neighbor of every odd q is divisible by four. This contradicts square-freeness.

For REF-007, the repeated-squaring exponents through $2^{t(k)}$ have divisor union exactly $\{2^j : 0 \leq j \leq t(k)\}$. At most two odd-prime candidates arise from each divisor, so the hit set is polynomial although $t(k) = \Theta(k \log k)$. BAR-003's exact pair bound makes the promised fraction vanish on every stipulated $2^{\Omega(k)}$ -size population. $\square$

J Reproduction appendix

The evidence implementation state preceding M6 is commit `ccc3a9368a015f9d4f1d840dda79d7a3c9a28ecf`. The M6 manuscript core commit is `9aac77921421fd69a4ab83e879cf50084f3024f3`; the final documentation commit is recorded in `research/STATUS.md`. The M7 nonsplit Lucas implementation and proof core is commit `171a3058c8a26b7d2e25641e8c35a9373c19bc06`. The M8 combined promise implementation and proof core is commit `bc8b25222823e06830530eac3962271c6d14a7ca`. The M9 exponent-encoding barrier implementation and proof core is commit `1416d70a6bc39a4c7491e9ec86e4e67f96293962`. The M10 multiplication straight-line compression implementation and proof core is commit `5501d2d1d2a6a5c584fdc03f905e9a36a6054733`. The M11 constant-sensitive boundary implementation and proof core is commit `cd391e5cf64207ea7dc2f6e4dd55cf469af45424`. The Windows AMD64 environment and exact toolchains are recorded in `research/toolchains/windows-amd64-20260725.json`.

The registered experiment cores and canonical summary hashes are:

- EXP-0002, core `7691b61dfef292d0da8af5eb022d62cc52383634`, hash `89bda0d3ea8054542151fda07d00c1e2711536b7339952618aea692c1d74cc59`;
- EXP-0003, core `93e97d96c544d5feddad208997834f47763cf31f`, hash `0a1d2ca2fef29126b60f3a9377454200e33fce20c0b49c081ea527622f8c536d`;
- EXP-0004, core `89cc44a6823ef223f36b37ad1cc268fe8fbd9697`, hash `4c046ae8694070b59f5e328f94038fe32cb84b5ab716bb86a62e79636077e55f`;

- EXP-0005, core 7ca79de49c404b10c1b1b355ff10b8a8d7d59635, hash 98f2be052a315231292c73319fa98066cf4d8fc4cd66740f207b2d99c7f616f5;
- EXP-0006, core 171a3058c8a26b7d2e25641e8c35a9373c19bc06, hash 23ed0067d2ccb642c3676ff4ea3f5c34e1e622f6372626aa84377eac74b7d905;
- EXP-0007, core bc8b25222823e06830530eac3962271c6d14a7ca, hash fb2f861f1670c3e4f68a0e8b461f430e7e10eeb966d9f5bec48886c810dd6cd3;
- EXP-0008, core 1416d70a6bc39a4c7491e9ec86e4e67f96293962, hash b8357f9436ef4d31d072f62dab4f3c8dedad41d6f1787803bf5df2f485ca53ed;
- EXP-0009, core 5501d2d1d2a6a5c584fdc03f905e9a36a6054733, hash 67508cf957fa356350a707a58f1079aebcea4f02481ff826cd5ed09727d210fa;
- EXP-0010, core cd391e5cf64207ea7dc2f6e4dd55cf469af45424, hash 22699f23a1421805cb472ddca1723e8d580d601cd25ca72b8b2cd134743e4f83.

The exact reproduction commands are:

```
python scripts/validate_foundation.py
python scripts/check_publication.py
python -m unittest discover -s tests -v
python -m compileall -q python scripts tests
cargo fmt --all --check
cargo clippy --workspace --all-targets --all-features -- -D warnings
cargo test --workspace --all-features
$mooseAppData = Join-Path (Get-Location) 'tmp/dotnet-appdata'
New-Item -ItemType Directory -Force $mooseAppData
$env:APPDATA = $mooseAppData
dotnet restore verification/csharp/MosefVerifier.csproj '
--configfile verification/csharp/NuGet.Config '
--packages .packages/nuget
dotnet build verification/csharp/MosefVerifier.csproj '
--no-restore --configuration Release
python scripts/check_baseline_differential.py
python scripts/run_m2_separator_search.py '
--n-max 500 --base-max 20 --exponent-max 20
python scripts/check_m2_separator_differential.py
python scripts/run_m3_semismooth_search.py '
--n-max 500 --base-bound 5 --smooth-bound 8 '
--cofactor-bound 3 --collision-bound-max 20
python scripts/check_m3_semismooth_differential.py
python scripts/run_m4_difference_cover_search.py '
--order-bound 8 --candidate-max 12 '
--modulus-max 200 --construction-bound 200
python scripts/check_m4_difference_cover_differential.py
python scripts/run_m5_multigroup_search.py '
--modulus-max 700 --base-max 32 '
--parameter-max 32 --exponent-max 12
```

```

python scripts/check_m5_multigroup_differential.py
python scripts/run_m7_nonsplit_search.py '
    --prime-max 43 --exponent-max 80
python scripts/check_m7_nonsplit_differential.py
python scripts/run_m8_promise_density_search.py '
    --prime-max 101 --candidate-max 18 '
    --family-size-max 3 --balanced-n-max 6
python scripts/check_m8_promise_density_differential.py
python scripts/run_m9_divisor_budget_search.py '
    --bit-length-max 18 --direct-exponent-max 4096 '
    --prime-max 4093
python scripts/check_m9_divisor_budget_differential.py
python scripts/run_m10_compressed_exponent_search.py '
    --step-max 7 --descriptor-level-max 16
python scripts/check_m10_compressed_exponent_differential.py
python scripts/run_m11_boundary_schedule_search.py '
    --exponent-max 262144 --primorial-count-max 12
python scripts/check_m11_boundary_schedule_differential.py
latexmk -xelatex -outdir=tmp/pdfs '
    -interaction=nonstopmode -halt-on-error paper/main.tex

```

The searches use no random seed. The randomized theorems are validated by exact successful-residue enumeration in EXP-0003 and exact parameter enumeration in EXP-0006 rather than by unrecorded random runs. EXP-0007 through EXP-0010 are deterministic combinatorial or syntax enumerations. Large raw datasets are not required; scripts, bounds, compact records, canonical vectors, commits, and hashes reconstruct the reported observations. The stable rendered artifact is `output/pdf/mosef-paper.pdf`.

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